Megan Ying

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Mechanical Engineering Experience

Apple *Interconnect Product Design Co-op* *Cupertino, CA | Sep 2023 – Aug 2024*

- Designed and integrated 4 connector systems for next-generation consumer devices in Siemens NX, meeting mechanical requirements and manufacturing constraints while reducing projected unit costs by up to 22%
- Drove production readiness by preparing 2D drawings and BOMs, reviewing Cpk results and first article inspections, and leading supplier and cross-functional DFM reviews
- Performed tolerance stackup analysis and FEA to validate connector fit across manufacturing tolerances and structural performance under worst-case loads
- Identified root causes of 2 connector failure modes through DOE and failure analysis, implemented design and assembly changes, and brought revised parts to production
- Automated analysis of 1,000+ Instron datasets using Python to accelerate design evaluation and identify parameters that minimized connector clearance while satisfying insertion-force requirements

Caterpillar *Hydraulics and Controls Engineering Intern* *Clayton, NC | May 2022 – Aug 2022*

- Evaluated hydraulic lift, lower, and tilt performance across small-wheel-loader operating modes, recommending control changes to reduce cycle times
- Developed Python and Power BI diagnostic tools to automate fault detection, identifying 8 high-priority machines for engineering investigation

Projects

Battery Door *Nov 2024 – Dec 2024*

- Designed a 9-component, one-touch battery-door release and re-latching mechanism in Siemens NX, using tolerance analysis and FEA to evaluate worst-case engagement and structural performance against a 1,000-cycle durability target

Trash Compactor *Sep 2024 – Dec 2024*

- Designed and fabricated an automated trash compactor in SolidWorks, integrating a linear actuator, stepper motor, machined scissor linkage, 3D-printed mounts, and Arduino controls

Tripod Assistive Technology *Nov 2022 – Dec 2022*

- Designed and prototyped an assistive tripod attachment in SolidWorks that automatically secures a phone using IR sensing and servo actuation, improving accessibility for users with limited hand strength

Additional Experience

Bloomberg *Software Engineer* *New York, NY | Oct 2025 – Present*

- Redesigned deployment processes for data pipelines supporting 250K+ securities, reducing manual steps by 40% while lowering deployment delays and risk of human error
- Identified and resolved a distributed-database capacity risk by redesigning the data architecture and verifying integrity across 28.3M migrated records

Design Research Collective *Research Assistant* *Pittsburgh, PA | Sep 2024 – Jul 2026*

- Published 2 first-author papers examining how reasoning strategy and model size affect AI agreement with expert engineering material-selection decisions
- Contributed simulated-annealing and reinforcement-learning modules to an open-source engineering design framework

UBS *Quantitative Trading Summer Analyst* *New York, NY | Jun 2023 – Aug 2023*

- Developed Python tools to evaluate pricing strategies and monitor risk across 1,000+ clients, enabling earlier loss detection and data-driven model selection

Education

Carnegie Mellon University (CMU) *Pittsburgh, PA | May 2025*

B.S. in *Mechanical Engineering* — GPA: 4.0

Skills

CAD & Design: Siemens NX, SolidWorks, GD&T, DFM | **Analysis:** Tolerance Analysis, FEA, ANSYS, DOE, Failure Analysis

Manufacturing: Injection Molding, Manual Machining, Stamping, 3D Printing | **Programming:** Python, Arduino, MATLAB, C